

Custom Exception Handling in Java

1. Aim

To create and use a custom exception in Java to check voting eligibility and handle an invalid age using a try-catch block.

2. Description

A custom exception is a user-defined exception created by extending the Exception class. In this program, InvalidAgeException is created for an age below 18. The checkVotingEligibility() method throws this exception when the given age is less than 18. The main() method catches the exception and displays the appropriate message.

3. Algorithm

1. Start the program.
2. Create InvalidAgeException by extending the Exception class.
3. Define checkVotingEligibility() with the throws InvalidAgeException declaration.
4. Check whether the given age is less than 18.
5. If the age is less than 18, throw InvalidAgeException.
6. Call checkVotingEligibility(16) inside the try block.
7. Catch InvalidAgeException and display its message.
8. End the program.

4. UML Diagram

InvalidAgeException	CustomException
+ InvalidAgeException(message : String)	+ checkVotingEligibility(age : int) : void + main(args : String[]) : void
extends Exception	

5. Source Code with Comments

```
class InvalidAgeException extends Exception {
    public InvalidAgeException(String message) {
        super(message);
    }
}

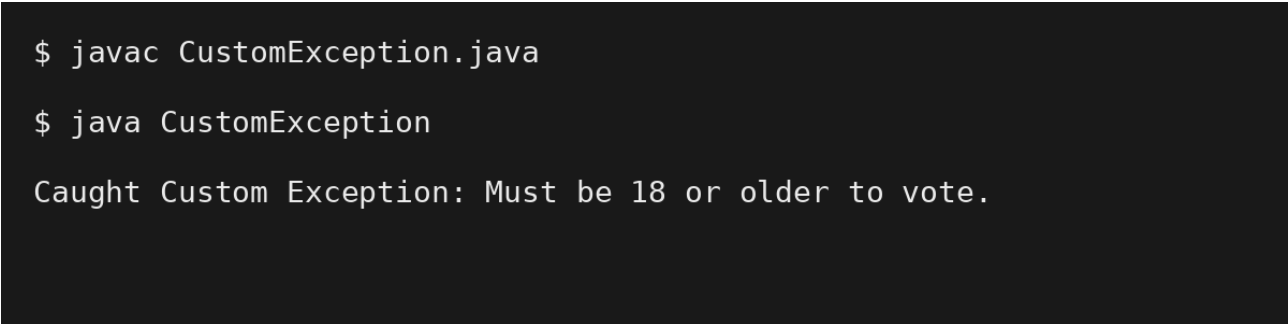
public class CustomException {
    public static void checkVotingEligibility(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException("Must be 18 or older to vote.");
        }
    }

    public static void main(String[] args) {
        try {
            checkVotingEligibility(16);
        } catch (InvalidAgeException e) {
            System.err.println("Caught Custom Exception: " + e.getMessage());
        }
    }
}
```

6. Compilation and Execution Commands

```
javac CustomException.java  
java CustomException
```

7. Output Screenshot



```
$ javac CustomException.java  
  
$ java CustomException  
  
Caught Custom Exception: Must be 18 or older to vote.
```

8. Result

The program was successfully compiled and executed. The custom `InvalidAgeException` was created, thrown for age 16, caught successfully, and the appropriate error message was displayed.